

18B.15

$$n_A z = -\rho D_{AB} \frac{dw_A}{dz}$$

~~$$\frac{dw_A}{dz} =$$~~

$$w_A = w_{A0}(z)$$

$$(n_A)_{\text{surface}} = f(w_{A0})$$

$$0 = \frac{S n_{Az}|_z - S n_{Az}|_{z+\Delta z} - \rho \Delta z f(w_A)}{S \Delta z}$$

$$n_A = \rho A V_A + j_A$$

$$0 = \frac{dn_{Az}}{dz} - \frac{\rho}{S} f(w_{A0})$$

b,

$$n_A = \rho V_A w_A + j_A$$

$$j_A = -\rho D_{AB} \nabla w_A$$

$$n_{Az} = -\rho D_{AB} \frac{dw_A}{dz}$$

$$n_A = -\rho D_{AB} \nabla w_A$$

$$c, \quad f(w_{A0}) = k_1'' w_{A0} \quad z=L$$

$$n_{Az} = -\rho D_{AB} \frac{dw_A}{dz}$$

$$-\frac{dn_{Az}}{dz} = \frac{\rho}{S} f(w_{A0})$$

$$-\frac{dn_{Az}}{dz} = \frac{\rho}{S} k_1'' w_{A0}$$

$$-\rho D_{AB} \frac{d^2 w_A}{dz^2} = \frac{\rho}{S} k_1'' w_{A0}$$

$$\begin{array}{l} z=0 \\ z=L \end{array} \quad \begin{array}{l} w_A = w_{A0} \\ n_{Az} = 0 \\ \frac{d^2 w_A}{dz^2} = 0 \end{array}$$

$$-\frac{d^2 \Gamma}{dz^2} - \phi^2 \Gamma = 0$$

$$\Gamma = C_1 \cosh \phi z + C_2 \sinh \phi z$$

$$F = \frac{\cosh \phi \cosh \phi z - \sinh \phi \sinh \phi z}{\cosh \phi}$$

$$\phi = \sqrt{\frac{\rho k_1'' L^2}{S D_{AB}}}$$

$$F = \frac{w_A}{w_{A0}}$$

$$z = \frac{z}{L}$$

$$\Gamma = \frac{\cosh [\varphi(1-\xi)]}{\cosh \varphi}$$

$$\frac{w_A}{w_{A1}} = \frac{\cosh \left[\sqrt{\frac{PL^2 k_1''}{SDAB}} (1-\xi) \right]}{\cosh \sqrt{\frac{PL^2 k_1''}{SDAB}}}$$

d.

$$\frac{w_{avg}}{w_{A1}} = \frac{\int_0^L \left(\frac{w_A}{w_{A1}} \right) dz}{\int_0^L dz} = \frac{\tanh \varphi}{\varphi}$$

$$n_{A2}/z=0 = -\rho DAB \frac{dw_A}{dz} \Big|_{z=0} = \left(\frac{\rho w_{A1} DAB}{L} \right) \varphi \tanh \varphi$$

e.

18 C3

$$W_A = \int_0^W \int_0^L N_{Ax} \Big|_{x=0} dz dy$$

$$W_A = \int_0^W \int_0^L C_{A0} \sqrt{\frac{DAB v_{max}}{\pi z}} dz dy$$

$$W_A = WL C_{A0} \sqrt{\frac{4 DAB v_{max}}{\pi L}}$$

$$\frac{C_A}{C_{A0}} = \operatorname{erfc} \frac{x}{\sqrt{4 DAB z / v_{max}}}$$

$$W_A = W v_{max} \int_0^\infty q \Big|_{z=L} dy$$

$$b. \quad C_A = C_{A0} \operatorname{erfc} \frac{x}{\sqrt{4 DAB z / v_{max}}}$$

$$W_A = W v_{max} \frac{C_{A0} L}{\sqrt{\pi}} \sqrt{\frac{4 DAB L}{v_{max}}} \int_0^\infty \left(\int_u^\infty \exp(-\xi^2) d\xi \right) du$$

$$u = \frac{x}{\sqrt{4 DAB z / v_{max}}}$$

19 B.1

a.

$$\frac{-k \left(\frac{dT}{dy} \right) \big|_{y=0}}{-k \left(\frac{dT}{dy} \right) \big|_{y=0}} = \frac{-(N_A \tilde{c}_p / k) \delta}{1 - \exp(-(N_A \tilde{c}_p / k) \delta)}$$

$$\sum_{\alpha=1}^N N_{\alpha} = c v^* \rightarrow N_A = c v^*$$

$$N_A = x_A (N_A + N_B) - c D_{AB} \nabla x_A = c v^* - c D_{AB} \nabla x_A$$

$$\sum_{\alpha=1}^N (c v^* - c D_{AB} \nabla x_A) = c v^*$$

$$v^* = \frac{N_A}{c}$$

$$v^* = \frac{N_A + c D_{AB} \nabla x_A}{c} = \frac{N_A + c D_{AB} \nabla x_A}{c x_A}$$

$$v^* = \frac{N_A}{c} = \frac{N_A + c D_{AB} \nabla x_A}{c}$$

b. $\frac{d^2 x_A}{dz^2} + \frac{1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 = 0$

19 D.1.

a. $\frac{d}{dt} \int_V \rho dV = \int_S (\mathbf{u} \cdot \mathbf{n}_A) dS + \int_V r dV$

$$\frac{\partial \rho_A}{\partial t} \Delta x \Delta y \Delta z = x y z n_{Ax} \big|_x - x y z n_{Ax} \big|_{x+\Delta x} + \Delta x \Delta y \Delta z r_A$$

$$\frac{\partial \rho_A}{\partial t} = -(\nabla \cdot \mathbf{n}_A) + r_A$$

$$\frac{\partial \rho_A}{\partial t} = - \left(\frac{\partial n_{Ax}}{\partial x} + \frac{\partial n_{Ay}}{\partial y} + \frac{\partial n_{Az}}{\partial z} \right) + r_A$$